Threshold drift and recalibration on BAF: one page

Question. A fraud model's threshold is set once at 5% FPR. How far do realized FPR, TPR and alert volume drift later, how unevenly across age groups, and which recalibration keeps the alert budget on target?

Setup. BAF (Jesus et al., NeurIPS 2022): a base dataset and five variants from one generator, so the six share one base and are not independent replications. LightGBM tuned on month 3 (20 configurations), trained once on months 0 to 2, 5 seeds; test months 4 to 7. That validation split is a single choice; other splits were run only post hoc. Strategies: static (threshold from month 3), label-free (top 5% of each month, or month 3's alert rate), delayed-label (previous labelled month), and an oracle on each month's own labels (an upper bound, not deployable). The primary comparison is budget miss, realized FPR minus 5%, because all strategies sit on the same month's ROC curve. Paired bootstrap, 1,000 resamples.

Finding 1: a threshold set once misses its budget from month 5 on, and the direction depends on the validation month. Set on month 3, static under-spends by 0.87 to 2.32 points in months 5 to 7 (18 of 18 intervals below zero, each 0.23 to 0.32 points wide, in six variants sharing one base). Post hoc, trained on months 0 and 1 with the threshold set on month 2, it over-spends by 0.33 to 5.82 points; with the model held fixed, a threshold set on month 5 over-spends in 9 of 12 cells. RESULTS.md amends the original "under-spends" claim.

Finding 2: the label-free results are largely arithmetic at this prevalence. Matching month 3's alert rate stays within 0.5 points in 22 of 24 variant-months, and top 5% of applications is under budget everywhere because frauds take part of the alerts. Since FPR = (alert share - TPR x prevalence) / (1 - prevalence), 5% FPR here means flagging 5.26% to 6.10% of applications; a fixed rule flagging 5.75%, chosen after seeing these test months, is within 0.5 points in 24 of 24.

Finding 3: recalibrating a global threshold moves the age-group FPR ratio much less than the variant does. Across global-threshold strategies the ratio differs by at most 0.094 in any variant-month. In Variant IV it stays between 0.081 and 0.125 before and after the prevalence disparity is removed, and in months 6 and 7 older applicants get about 89% of label-free top 5% alerts at 51% of applications. Group-wise thresholds (exploratory, using age) give ratios of 0.547 to 0.998.

Limits. The shuffled-label check failed its declared tolerance (pooled TPR 0.117 against 0.05 plus or minus 0.02); no leak was found and the median of 20 repeat shuffles was inside the tolerance, but the cause is not established; it stays a caveat. BAF is synthetic: per the datasheet (answers A6, p1, and A14, p4) and the Kaggle page, it was sampled from a CTGAN trained on noise-perturbed bank data, so month-to-month shifts may reflect the generator rather than applicant behaviour. One main split, no retraining in the main study; no single feature among the ten most important explains the static miss. The contribution is a measurement on BAF and a month-by-month strategy comparison by budget miss, not the observation that thresholds drift.